

Thapar Institute of Engineering & Technology, Patiala
(Deemed to be University)
Department of Electronics & Communication Engineering
END SEMESTER TEST (Summer Semester) July-2025

Roll Number: _____

Group: _____

B. Tech: ECE, ENC, EVD

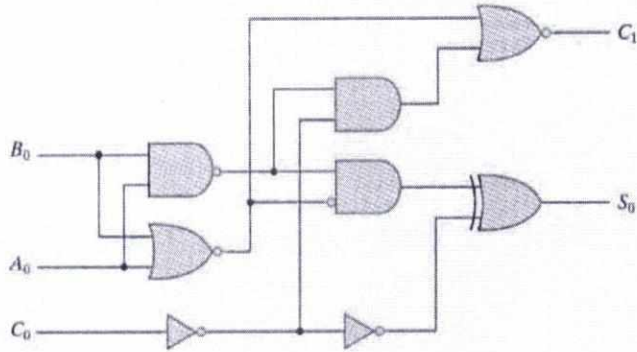
Date- 19.07.2025

Time: 03 Hours, M.M.: 40

UEC612: Digital System Design

Name of Instructor: Dr. Amit Kumar

NOTE: * Attempt any 5 questions and all parts of a question in one place. Otherwise, only the first attempt will be evaluated. Make suitable assumptions, if required.

Q.	Questions	Marks	CO	BL
Q1	Simplify the given Boolean functions using the Q-M method. $F(A,B,C,D,E,F) = \sum m(6,7,8,9) + d(10,11,12,13,14,15)$	8	CO 1	L3
Q2	a) Implement an active high-output 1:8 Demultiplexer using combinational gates. b) Define the carry propagate and carry generate: $P_i = A_i + B_i$ $G_i = A_i B_i$ respectively. Show that the output carry and output sum of a full adder become: $C_{i+1} = (C_i G_i + P_i)$ $S_i = (P_i G_i) \oplus C_i$ The logic diagram of the first stage of a four-bit parallel adder as implemented in IC type 74283 is shown in the figure below. Identify the P_i and G_i terminals and show that the circuit implements a full adder. 	4 + 4	CO 1	L4
Q3	a) Design a combinatorial circuit using ROM. The circuit accepts a 3-bit binary number and outputs a binary number equal to the square of the input number. b) Implement $f_1(a, b, c) = m(3, 5, 6, 7)$ and $f_2(a, b, c) = m(0, 2, 4)$ with a PLA.	4 + 4	CO 2	L4
Q4	Convert by adding external gates: a) a D flip-flop to a J-K flip-flop. b) a T flip-flop to a D flip-flop with clock enable.	4 + 4	CO 3	L5

Q5	An AB latch operates as follows: If $A = 0$ and $B = 0$, the latch state is $Q = 0$; if either $A = 1$ or $B = 1$ (but not both), the latch output does not change; and when both $A = 1$ and $B = 1$, the latch state is $Q = 1$. a) Construct the state table and derive the characteristic equation for this AB latch. b) Derive a circuit for the AB latch that has four two-input NAND gates and two inverters. c) Derive a circuit for the AB latch using four two-input NOR gates and two inverters.	4 + 2 + 2	CO 3	L5
Q6	a) Design a shift register that can multiply and divide the number by 2^n through the same circuit. b) Draw the state diagram and state table of a Moore-type sequence detector to detect the sequence 110.	4 + 4	CO 4	L5

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